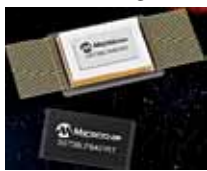


COMPONENTS

Radiation-tolerant memory

Microchip's SST38LF6401RT is a radiation-tolerant, 64Mbit parallel-interface SuperFlash memory device. It has total ionizing dose radiation tolerance



of up to 50krad for reliability and robustness in the harsh space environment, which reduces the time, cost, and risk of developing spaceflight-qualified systems. It allows designers to start with commercial off-the-shelf devices that can later be replaced by their space-qualified, radiation-tolerant equivalent parts.

Microchip Technology
www.microchip.com

High-speed op-amp

The 2-channel, high-speed, ground sense CMOS op-amp BD77502FVM from ROHM Semiconductor mitigates



noise issues associated with the large number of electronic components used in an IoT environment.

The noisy environment makes it extremely difficult to implement designs with electromagnetic compatibility. This op-amp is optimised for consumer and industrial equipment requiring high-speed sensing for industrial measurement and control systems.

ROHM Semiconductor
www.rohm.com

LoRa compatible SoC

The STM32WL long-range sub-GHz wireless system-on-chip family by STMicroelectronics enables connection with LoRa-based low-power wide-area networks. It allows users to create com-

pact, energy-efficient, and reliable IoT devices for cost-effective connections over large geographical areas and remote



locations. This enables smart technologies to add greater value in industries from utilities and agriculture to shipping, transportation, and others. The STM32WL series supports multiple modulation schemes, which allow high receiving sensitivity at low RF signal power, and (G)FSK, (G)MSK, and BPSK modulation used by Sigfox and wireless Meter-Bus protocols.

STMicroelectronics
www.st.com

Solar for wearables and IoT

The tradeoff between small size and long runtime for wearable and IoT applications often poses a challenge for space-constrained designs. Reducing this problem is the MAX20361 single/



multi-cell solar harvester from Maxim Integrated, which has maximum power point tracking

capability. By enabling solar charging in highly space-constrained products, the MAX20361 extends device runtime with a supplemental power source, reducing solution size by at least half while boosting the efficiency of harvested energy by up to 5 per cent, making it ideal for wearables and other IoT applications.

Maxim Integrated
www.maximintegrated.com

Indoor air quality sensor

The ZMOD4410 indoor air quality sensor from Renesas Electronics can operate in wet or dirty environments. It has a software-configurable IP67-qualified

waterproof option for smart HVAC systems, ventilator fans, kitchen and bathroom lights, and switches and wearables, that operate in high-humidity, splash zone, dirty, or submersible environ-



ments. The IP67-rated packaging enables accuracy and reliability for low-power applications that operate in splash zones. It

maintains the high precision and flexibility that users need for customising their systems and eliminate the need for expensive waterproofing systems.

Renesas Electronics Corporation
www.renesas.com

High-power UV-C LED

UV-C light can eliminate most viruses and bacteria on surfaces, in the air, and in drinking water—making them harmless and preventing them from spreading. UV-C light has often been used with



large lamps for disinfection purposes. Now, with the Osilon UV 3636, UV-C light can be obtained with the help of LEDs for

small disinfection applications. The LED operates in a wavelength of 275 nanometers and poses no risk to human beings. Thus, it can be safely installed for direct interaction with the objects being sanitised such as faucets and air-conditioning systems.

OSRAM
www.osram.com

Power monitoring chip

The ACS37800, a Hall-effect power monitoring IC from Allegro MicroSystems, allows simultaneous measurement of power, voltage, and current for AC



and DC signals with an isolation rating of up to 1480Vpk in a small SOIC16